ANANT VARMA

AI/ML Engineer | Multi-Agent Systems | Generative AI

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PROFESSIONAL SUMMARY

Final year B.Tech Cybersecurity student with 8+ months of industry experience architecting multi-agent AI systems and LLM-powered platforms across two U.S. remote internships. Co-inventor on **6 granted patents and registered designs** (India & UK) across AI, IoT, and cybersecurity. Published an open-source Python security toolkit to PyPI. Seeking an AI/ML internship to apply hands-on skills in agentic workflows, RAG pipelines, and production-grade generative AI engineering.

PATENTS & INTELLECTUAL PROPERTY

Co-inventor on 2 granted patents (India) and 4 registered designs (UK & India):

- Intelligent Security-Aware 6G-IoT Architecture for Smart Cities (Indian Patent 202631008090)*Mar 2026*

-AI-Based Sustainable Agriculture Models for Disease Prevention (Indian Patent 202641008089)*Feb 2026*

-Computing Device for Food Supply Chain Optimization (Indian Design 488379-001)*Mar 2026*

-Quantum-Based Intelligent Testing Device for Agriculture Monitoring (UK Design 6505161)*Feb 2026*

-AI-Powered ERP Kiosk for Decision Support (UK Design 6502040)*Jan 2026*

-AI-Based Smart Forest Fire Detection System (UK Design 6502028)*Jan 2026*

PROFESSIONAL EXPERIENCE

DataObserve — Houston, TX (Remote)*Feb 2026 – Apr 2026*

AI/ML Engineer Intern

-Architected an AI agent ecosystem managing frontend, database, and API layers, reducing ML pipeline failure rate by ~30% through systematic edge-case testing covering 15+ failure scenarios.

-Engineered algorithmic trading logic in a multi-agent LLM/SLM framework; improved signal accuracy by 15% by integrating multi-layer noise filtration with a centralized data source-of-truth.

-Embedded structured user feedback loops into the agent development cycle, reducing iteration cycles by 2x and improving end-to-end reliability across agent handoffs.

exera AI — West Des Moines, IA (Remote)*Dec 2025 – Feb 2026*

Machine Learning Engineer Intern

-Built OlympusOS, an enterprise multi-agent AI chatbot platform orchestrating 7 parallel specialized agents via LangGraph, reducing agent response latency by ~80% through optimized context management. -Refactored agentic flows and prompt pipelines, cutting API overhead by eliminating redundant LLM calls and improving context window utilization across conversational agents.

-Designed and deployed document ingestion pipelines supporting PDF, DOCX, and structured data extraction, accelerating downstream analysis workflows for internal QA teams.

KEY PROJECTS

Sentinel-X | *Python, LangGraph, Claude/GPT, Nmap, ClamAV, Docker*

-Engineered a local-first cybersecurity platform integrating a 6-agent AI system with real-world security tools (Nmap, ClamAV, YARA); packaged and published as an open-source Python library on PyPI.

-Built a multi-provider LLM layer with streaming, session memory, and provider failover — supporting Claude, GPT, and local models with zero-downtime switching.

Advanced Network Scanner | *Python, Scapy, Dash, Plotly, Docker*

-Engineered a multithreaded real-time scanning tool using ICMP and TCP SYN protocols with dynamic Dash-based topology visualizations, achieving a 40% reduction in scan latency vs. sequential methods.

-Containerized the full stack via Docker Compose with real-time health monitoring and automated network topology rendering for up to 500-node environments.

AI Support Desk | *Python, FastAPI, React, Gemini, Zammad, Docker*

-Developed a full-stack platform where an AI agent triages support issues, auto-classifies severity, and generates tickets via Zammad REST API — reducing manual triage time by an estimated 60%.

-Deployed via Docker Compose with real-time health monitoring; achieved sub-200ms API response times for ticket creation at scale.

AI-Driven Trading Signal Platform | *Python, FastAPI, Next.js 14, Pandas*

-Built a quantitative trading signal platform featuring a crossover-based strategy engine with multi-layer noise filtration, automated 5-year historical backfilling, and a Next.js dashboard for signal visualization.

-Implemented a backend-as-source-of-truth architecture ensuring deterministic signal replay and auditability across all backtested strategies.

EDUCATION

Vellore Institute of Technology — Bhopal, India*Aug 2023 – Jun 2027 (Expected)*

B.Tech, Computer Science Engineering (Cybersecurity)

TECHNICAL SKILLS

AI / ML: LangGraph, Google ADK, Multi-Agent Systems, RAG, Vector Databases, Embeddings, Prompt Engineering, PyTorch, TensorFlow, NLP, Agentic AI, n8n

Languages: Python, JavaScript, TypeScript, SQL, Solidity, C++ (Arduino)

Frameworks: FastAPI, React, Next.js 14/15, Express.js, Dash, Plotly, TailwindCSS

Databases: PostgreSQL, SQLite, Redis, Elasticsearch, Firebase, AstraDB (Vector DB), Prisma ORM

DevOps: Docker, Docker Compose, nginx, Git, GitHub Actions

Cybersecurity: Nmap, ClamAV, YARA, Burp Suite, Metasploit, Scapy

CERTIFICATIONS

-Google Cybersecurity Professional Certificate — Coursera

-Google IT Support Certificate – Google